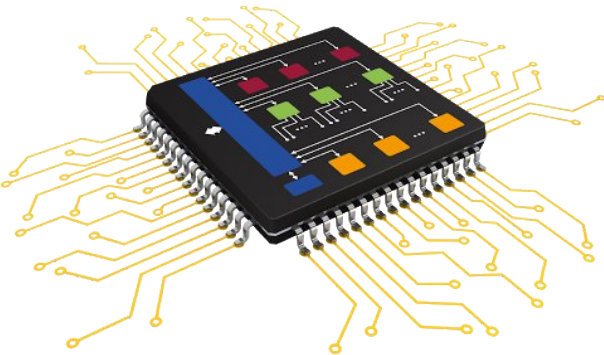


MODULE - 3

ARCHITECTURE OF SPECIAL PURPOSE COMPUTING SYSTEM

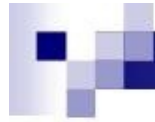


MODULE - 3

Architecture of Special Purpose Computing System

ATM, Handheld devices, Data Compressor, Image Capturing Devices Architecture and Requirements, Challenges Constraints of special purpose computing system.

Handheld devices



Handheld Definition

- Extremely portable, self-contained information management and communication devices.
- Small size and light weight, often made up of a single component.
- Sacrifices for mobility are less memory and little or no storage, small displays, and slow, unreliable access to the Internet.

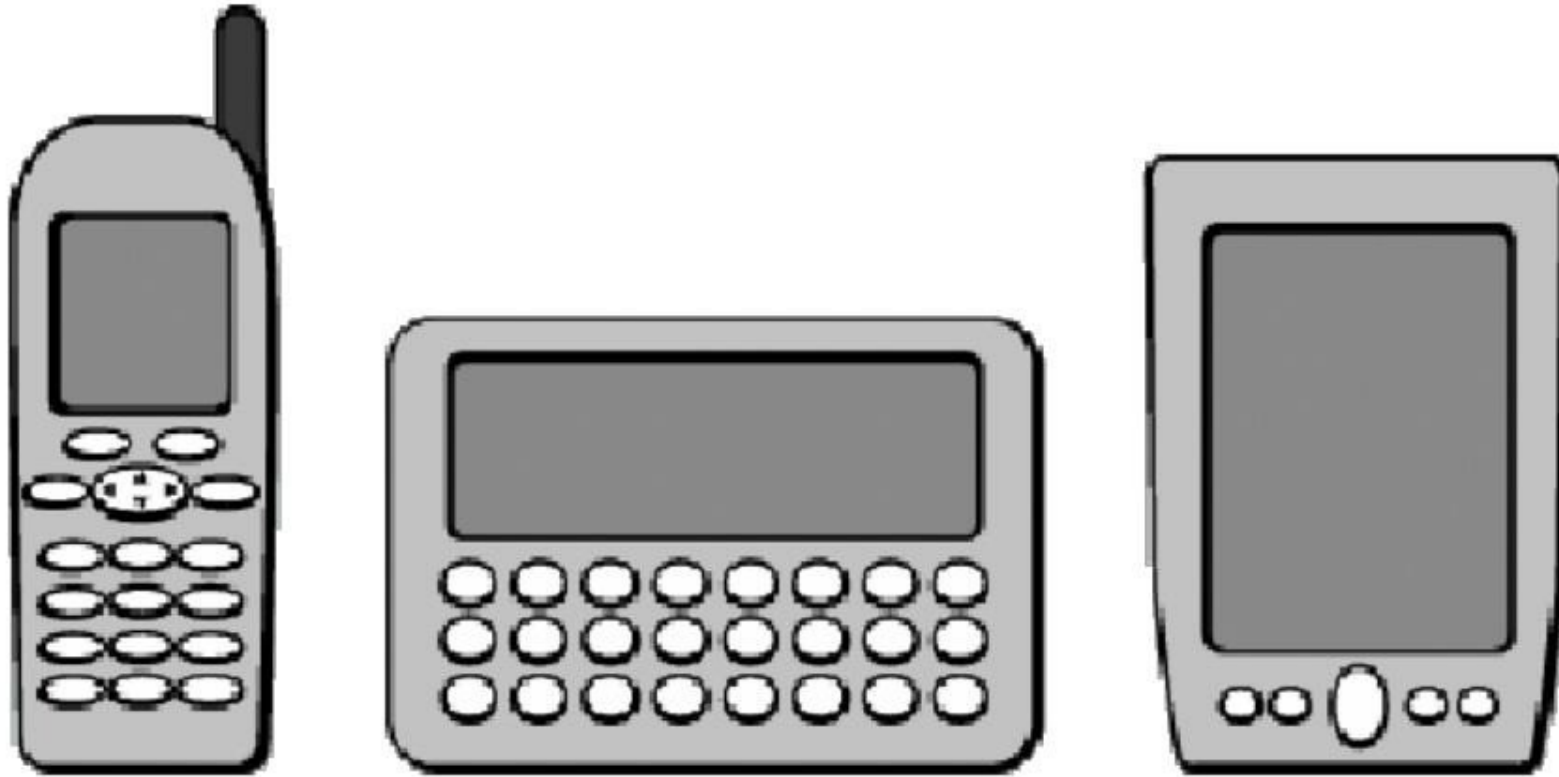


Handheld Definition

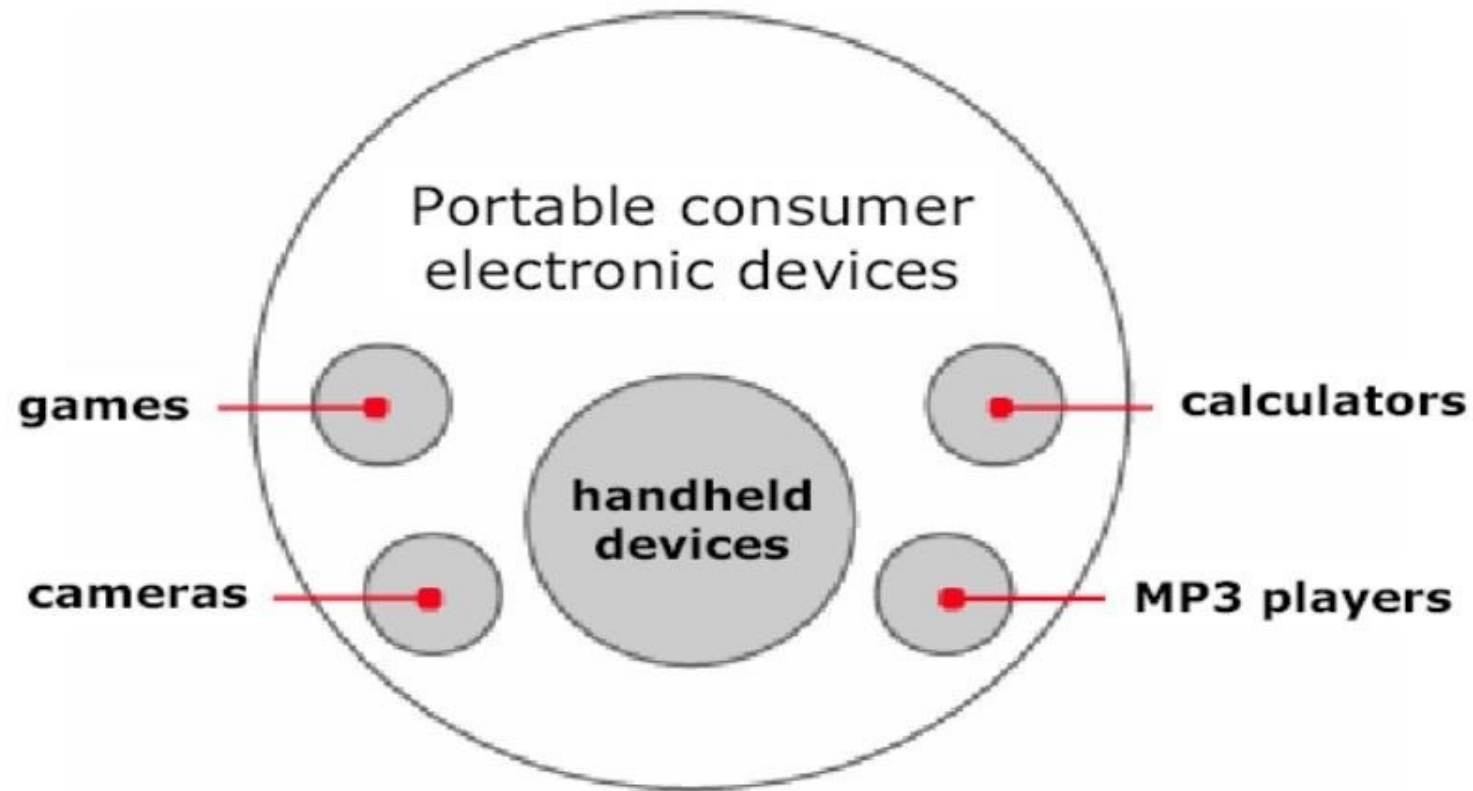
In addition, handheld devices must be able to

- Operate without cables, (except temporarily recharging, synchronizing with a desktop).
- Be easily used while in one's hands
- Must allow the addition of applications or support Internet Connectivity.

Handheld Devices



What is Not A Handheld?



Types of Portable Consumer Electronics Devices



Handheld Vs. Desktop

Handheld

- On-the-go lookup and entry of information.
- Form Factor : less than 10" x 12"
- Mobility : Can be used while on the go
- Connectivity: Slow and unreliable
- Input: Stylus, keypad, mini-keyboard, roller-wheel.
- Display Size: 240 x 320 pixels or less.
- Memory: Up to 16 megabytes.
- Storage: Sometimes none. Removable memory stick.



Handheld Vs. Desktop

Desktop

- Lengthy information processing tasks.
- Form Factor: Requires a table.
- Mobility: Requires a power cord, or frequent recharges
- Connectivity: Fast and reliable.
- Input: Full keyboard and mouse.
- Display size: 640 x 480 pixels or more.
- Memory: Up to 1 gigabyte.
- Storage: Diskettes, CDRW, DVD RAM rewritable media.



Types of Handheld Devices

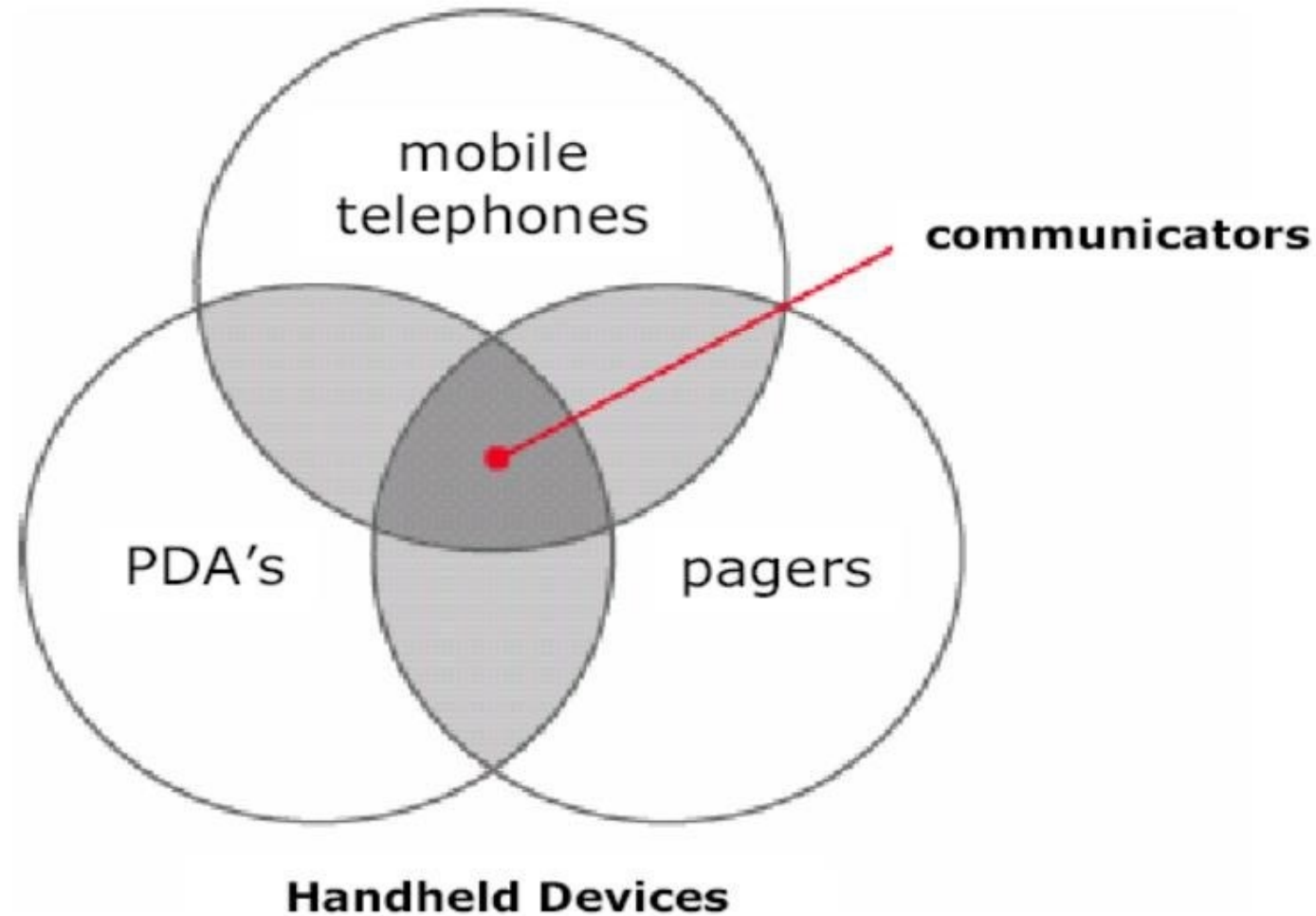
There are three essential types of handheld devices:

- Mobile telephone handsets
- Pagers
- PDAs

All handheld devices share two common challenges:

- Small display
- Problematic data input.

Types of Devices





Handheld Phones

- Primary use - Voice communication
- Data connectivity via WAP (Wireless Application Protocol) or i-mode.
- Data is entered on a standard twelve-key keypad, with the help of various soft keys.
- Bitmap graphics or user interface 'widgets' such as menus, text entry fields are not supported in WAP implementations.
- WAP applications are also extremely inconsistent with each other.



PDA's

- Primary use – Information storage and retrieval.
- Support address books, calendars, to-do lists, email management, and note taking.
- Have touch screens and use a stylus for input.
- User Interface Environments – Palm OS, Windows CE



Pagers

- Primary use – two-way email communication.
- Also support Web browsing.
- Have QWERTY keypads.
- User Interface Environments – RIM OS, Wisdom OS.
- RIM OS – simple graphical design, mostly text and icons.
- Wisdom OS – rich graphical interface supporting graphical controls such as radio buttons and check boxes.

User Interface Design Guidelines for Handheld Devices

1. 'Select' Vs. 'Type'

- Typing on handheld devices is difficult.
- Alternative input methods - Graffiti handwriting recognition software and on-screen keyboards.
- So when possible, offer a selection mechanism rather than requiring typing.





Design Guidelines

2. Design for Small Screen

- Small screens are difficult to use.
- A fundamental requirement is that content be contiguous without blank lines.

3. Consistency

- Use of same terminology and interaction schema within the same application and between applications reduces the learning curve for new features.



Design Guidelines

4. Design Stability

- Wireless data connections are prone to failure.
- So it is critical that the applications provide a stable user interface.
- ‘Stability’ implies that when the network connection drops, the application should restore state and context once the network goes back online.
- The network could be unstable, but the design need not be.



Design Guidelines

5. Providing Feedback

- Each page of an application should provide the user with enough information to understand what the application is and how to navigate from the page.

6. Forgiveness

- If a user makes a mistake, the user interface must offer means to correct it.
- Use 'Back' button or 'Undo' button.



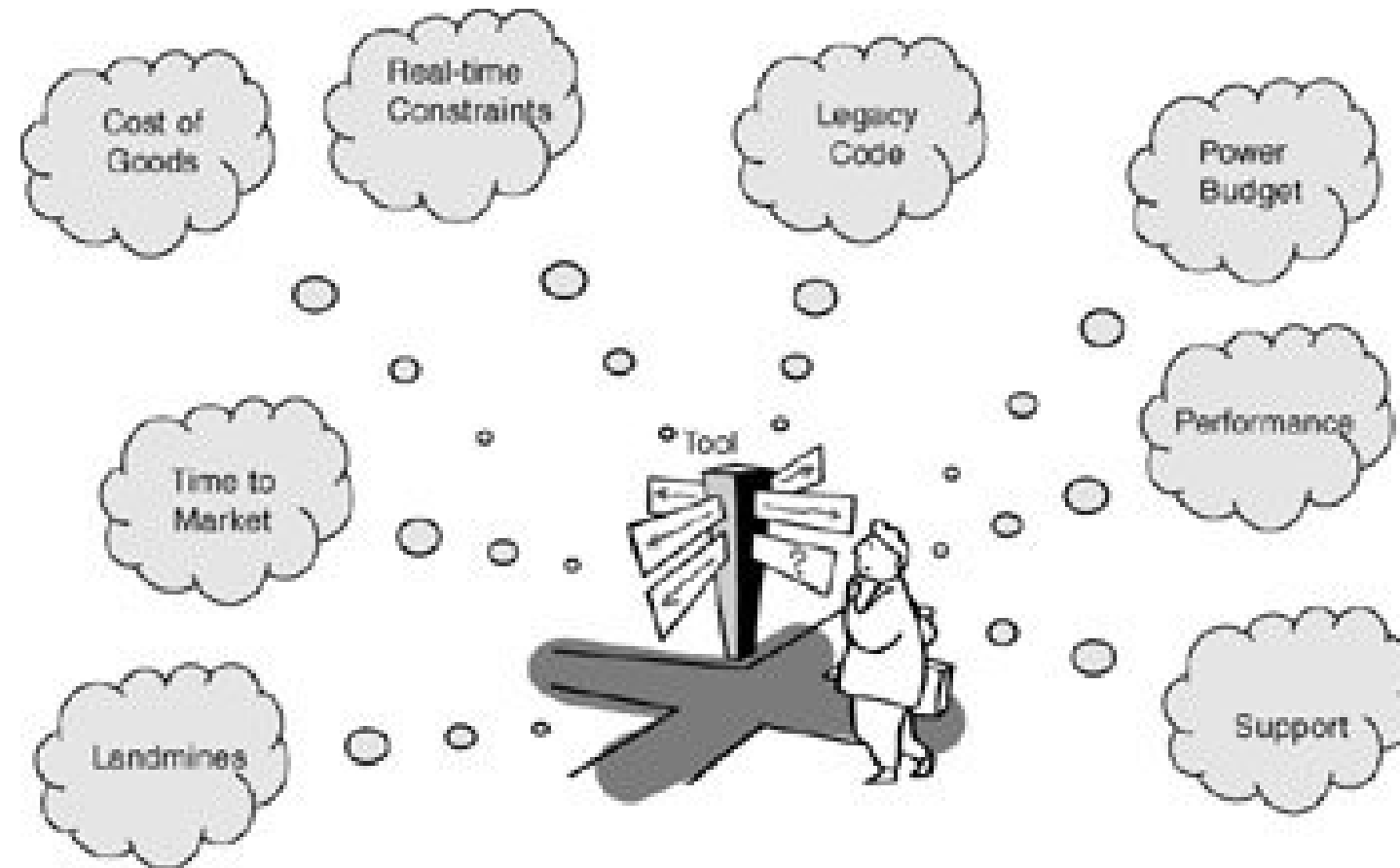
Design Guidelines

7. Use Metaphors

- Use metaphors from the real world.
- 'Bookmark' is a very effective metaphor for the wireless Web.

STRATEGIC SELECTION OF PROCESSOR & MEMORY

Selecting The Right Processor



Selecting The Right Processor

- ❑ Is it available in a suitable implementation?
 - ❑ What good is choosing the highest performing processor if the cost of goods makes your product noncompetitive in the marketplace?
- ❑ Is the Processor Capable of Sufficient Performance?
 - ❑ Processor must be able to do the job on time.

Selecting The Right Processor

- ❑ Is the Processor Supported by an Appropriate Operating System?
 - ❑ Porting the RTOS kernel to a new or different microprocessor architecture and having it specifically optimized to take advantage of the low-level performance features of that microprocessor is not a task for the faint-hearted.
- ❑ Is the Processor Supported by Appropriate and Adequate Tools?
 - ❑ Good tools are critical to project success.
 - ❑ At a minimum, you'll need a good crosscompiler and good debugging support.

PROCESSOR SELECTION CRITERIA

- ❑ To design an efficient embedded system, selection of right processor is very important and challenging task
- ❑ Types of processors: μ P, μ C, Digital signal processor (DSP)
- ❑ μ P are offered in 4 to 64-bit size with distinct features like cost, speed, no. of CPU core, address & data line are used in simple toys to network router
- ❑ μ C plays an important role in embedded system design and majorly used in low-end to high-end control applications
- ❑ DSP are majorly used for high computation intensive applications such as image processing, communication devices, voice to text converter etc.,

PROCESSOR SELECTION CRITERIA

- ❑ Sequence of analysis to be made selecting an appropriate processor for embedded system applications as follows,
 1. Application requirement analysis: understand the purpose of application and arrive specific requirement
 2. Processor Architecture analysis: MCS51, ARM, PIC, PowerPC, MIPS etc.,
 3. Peripheral set analysis: Includes on-chip (RAM, ROM ,I/O Ports, ADC) and specialized processing units (FPU, MMU, DMA)
 4. Technical analysis: Execution speed, operating voltage, power consumption, and data & address bus size etc.,
 5. Non-technical analysis: Cost, software tools, package type, vendor reputation, support etc.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Wireless Spot Billing Machine

- ❑ POS Hand Held spot Billing Machine (SBM) is a GPRS Enabled machine. The device is compact and lightweight(less than 500grms).
- ❑ The machine is equipped with technology which serves as a Hand Held computer.
- ❑ The device is WEB/USB enabled which helps the operator to get instant bill remotely and update transactions back to the server.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Wireless Spot Billing Machine



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

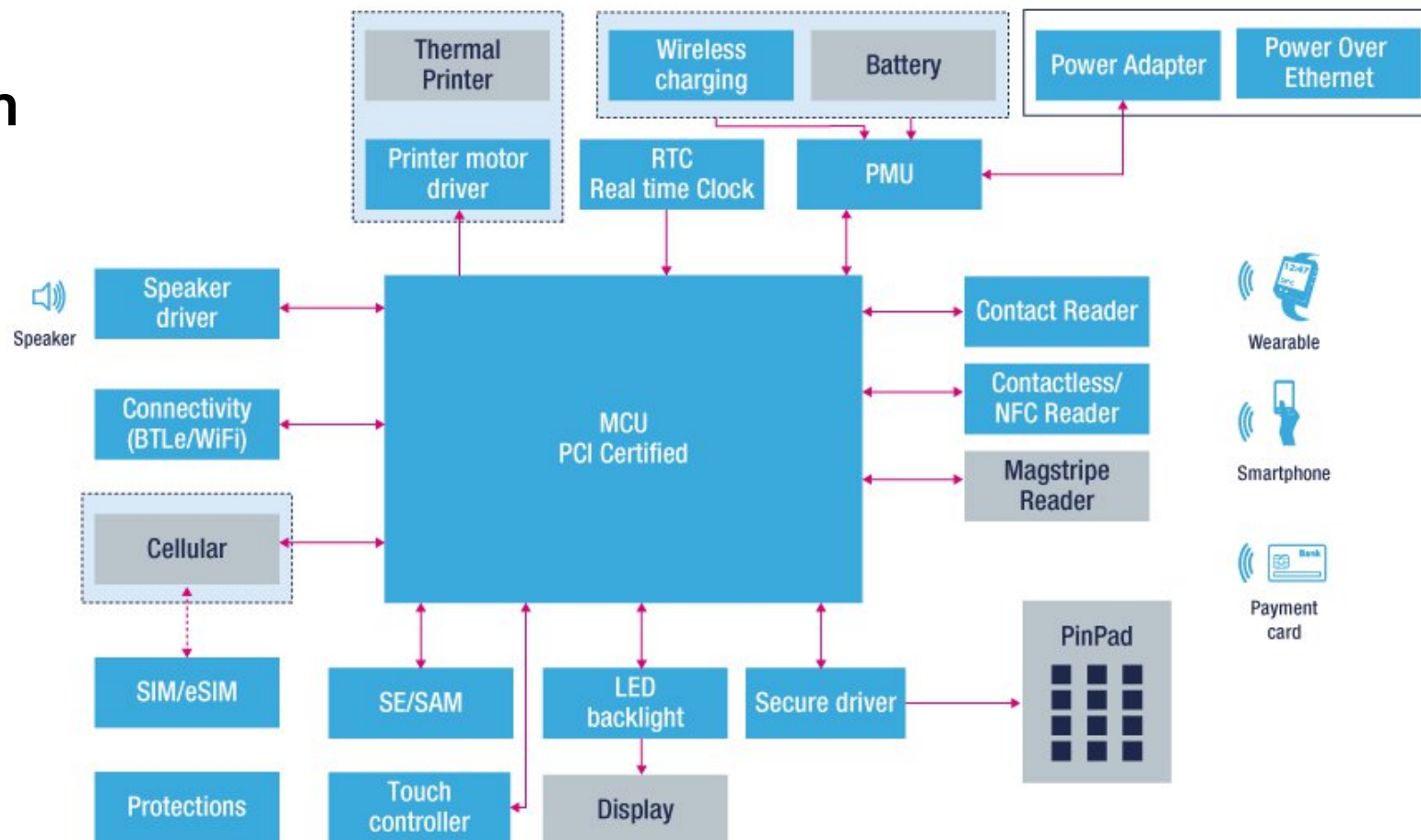
Application Spot Billing Machine spot billing machine

- ☐ Electricity and Water Billing
- ☐ Cable and Internet Billing
- ☐ Canteen and Hotel Billing
- ☐ Finance and Daily Bill Collection
- ☐ Bus and Parking Ticketing
- ☐ Distribution and Retail Billing
- ☐ Petrol and Toll Gate Tokens
- ☐ All kinds of General Billing

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Block Diagram



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Features of Spot Billing Machine spot billing machine

- ☐ Billing Machine is easy to operate and user-friendly.
- ☐ It is highly secured with Password protection.
- ☐ It is accessible to the remote application via GPRS.
- ☐ It holds the Customer Data in memory.
- ☐ Accurate data Import and Export Settings

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Specification

- Processor : ARM 32-bit Cortex-M3 CPU(120 Mhz max)with Adaptive real-time accelerator (ART Accelerator), MPU,150DMIPS/1.25DMIPS/MHz(Dhrystone 2.1)capable of In-System Programming.
On The Fly Programming(Can be done in the field using laptop)for Upgrading the units through Serial/USB Ports.
Choice of Program Memory Capacity of 512K/1024K bytes.
- Memory : 8 MB of Non-volatile data memory is provided with the units data retention is minimum 10 years with out any power being applied. Expandable to 16 MB
Optional - It also has micro SD card slat for large memory backup.
- Real Time Clock : In-built Real time clock with battery backup.
- Key-Board : A 35 key multi function key board is provided on the front panel.
- LC Display : 132*64 pixels graphical LCD display with back light provided for user interaction. It has various graphical icons for indication of battery status, signal status etc.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Specification

- Printer : High Speed 24 Column Impact Printer With 2.7 Lines/Sec, is Provided as Standard fitment.
- Paper Roll : 57.5mm+/-0.5mm,60mm dia Paper Roll Fitment.
- Communication : a) RS232C Ports-1 Nos(Option of second port)with flexible baud rates.
b) RS232C Ports-1 Nos(Option of second port)with flexible baud rates.
c) Optional IR or IRDA or Both IR or IRDA Port for Communication with electricity meters.
d) Built-in GSM/GPRS modem optional.
e) Optional GPS.
- Smart Card Interface : Optional Contact & Contact less(MIFARE)Smart Card Interface.
- Batteries : The units are powered by 7.4v 2600 mAH LI-ION Rechargeable Battery pack.
- Battery Charger : External Universal voltage AC/DC 9.7V-2Amp Adaptor provided.
- Mechanical Dimension : Width-102.00,Length-276.00,Height-82.00.
- Weight : Weight-655grams~(without paper roll).

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

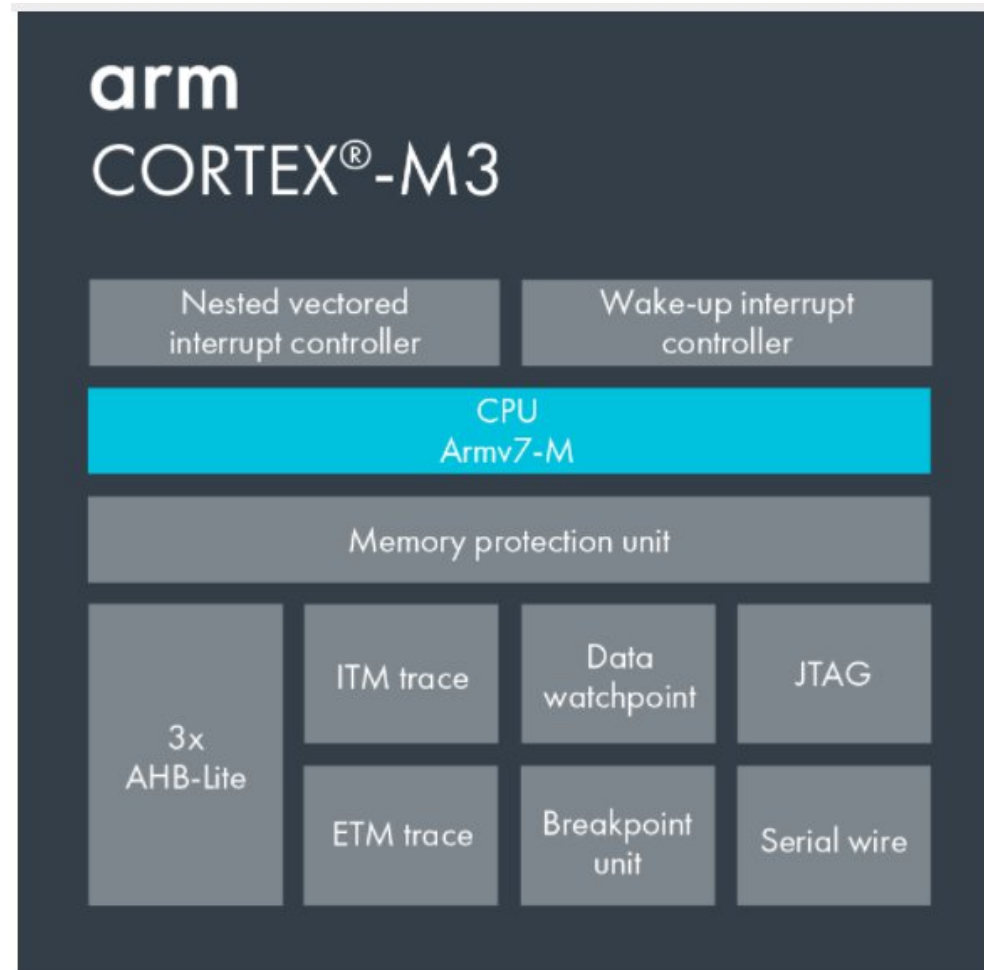
Processor

The Cortex-M3 processor is specifically developed for high-performance, low-cost platforms for a broad range of devices including microcontrollers, automotive body systems, industrial control systems and wireless networking and sensors

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Processor



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

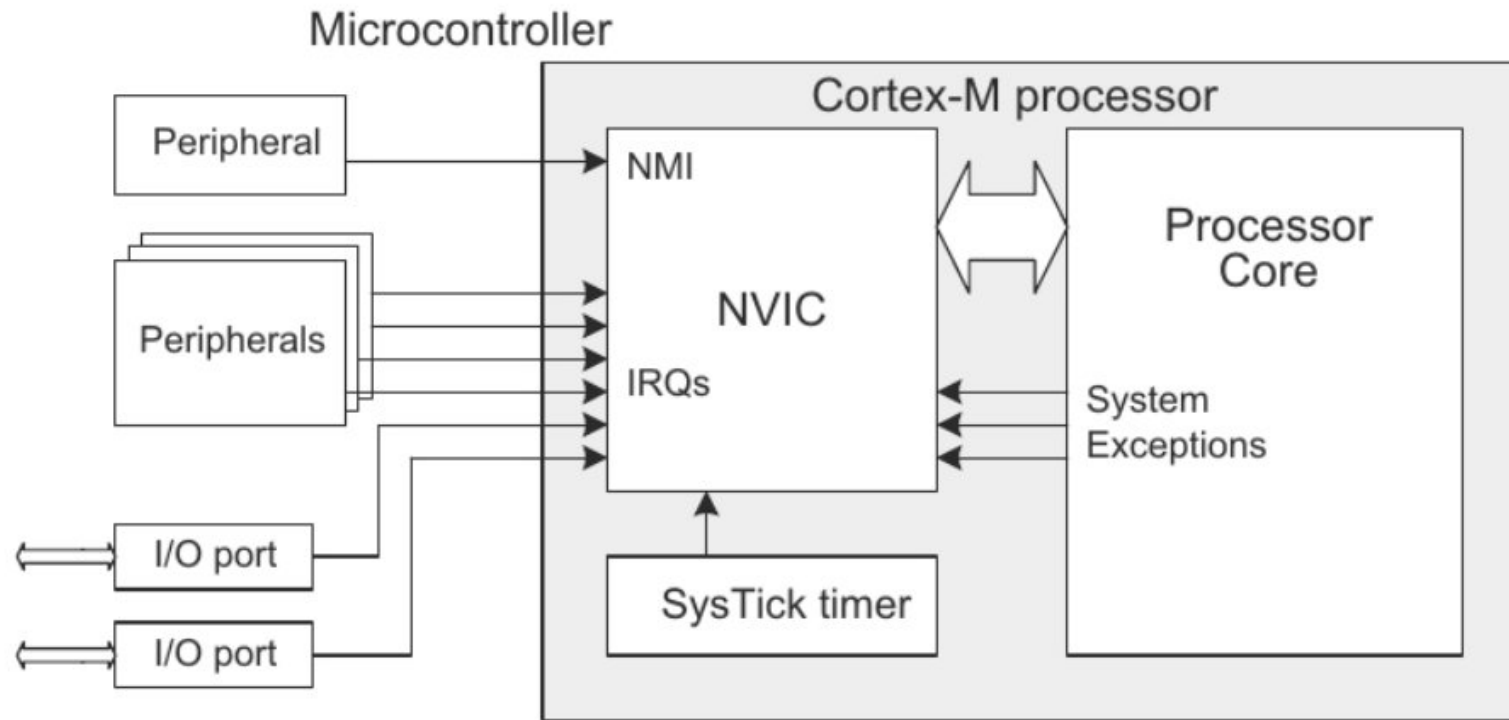
Processor

Architecture	Armv7-M
Bus Interface	3x AMBA AHB-Lite interface (Harvard bus architecture) AMBA ATB interface for CoreSight debug components
ISA Support	Thumb/Thumb-2 subset
Pipeline	Three-stage
Memory Protection	Optional 8 region MPU with sub regions and background region
Bit Manipulation	Integrated Bit-field Processing Instructions and Bus Level Bit Banding
Interrupts	Non-maskable Interrupt (NMI) + 1 to 240 physical interrupts
Interrupt Priority Levels	8 to 256 priority levels
Wake-up Interrupt Controller	Optional
Enhanced Instructions	Hardware Divide (2-12 Cycles), Single-Cycle (32x32) Multiply, Saturated Adjustment Support
Sleep Modes	Integrated WFI and WFE Instructions and Sleep On Exit capability. Sleep and Deep Sleep Signals Optional Retention Mode with Arm Power Management Kit

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Nested Vector Interrupt



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Nested Vector Interrupt

- ❑ Nested vector interrupt control (NVIC) is a method of prioritizing interrupts, improving the MCU's performance and reducing interrupt latency.
- ❑ NVIC also provides implementation schemes for handling interrupts that occur when other interrupts are being executed or when the CPU is in the process of restoring its previous state and resuming its suspended process.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Rugged Handheld Data Terminal

- ❑ It is built for life out in the field, integrates Corning Gorilla glass for display, able to handle drops, bumps, spills, dust, vibration.
- ❑ With stable wireless connections and accurate efficient data capture options, you can find this easy-to-deploy device a valuable helper to increase productivity in logistics, express delivery, warehousing, retail and industrial manufacture etc.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Rugged Handheld Data Terminal

- ☐ **Android 10.0 OS**
- ☐ **8GB ROM + 1 GB RAM (16GB+2GB Optional)**
- ☐ **4" high resolution(480*800) IPS TFT display**
- ☐ **8MP AF Camera with LED flash**
- ☐ **2G,3G,4G ,WiFi, Bluetooth connection**
- ☐ **IP 65 Sealing**
- ☐ **2D Zebra SE4710 barcode scanner**



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Specification

CPU	Qualcomm Quad-core ARM Cortex-A53 1.3 GHz processor
Operating System	Android 10.0 OS
Memory	8GB ROM + 1 GB RAM(16GB+2GB Optional)
SIM Slot	Support Micro SIM card
PSAM Slot	Support PSAM card (SIM 2 card Optional)
Expansion Slot	Micro SD card (Up to 128G)
Display	4.0 inch high resolution (480*800) IPS TFT display
Touch Panel	Industrial-grade capacitive touch panel (supporting operation with gloves)
Camera	8 Mega-Pixel Auto focus with LED flash
Scanner window	Tempered glass
Wireless LAN	Wi-Fi 802.11a/b/g/n/ac ; 2.4G/5G

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

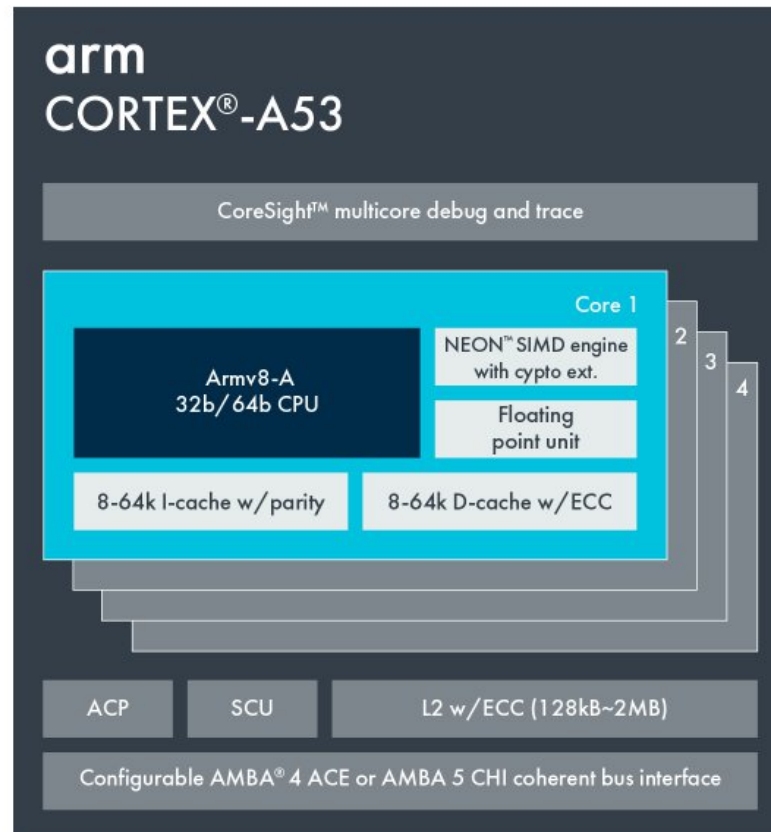
Processor

- ❑ The Cortex-A53 processor is a high efficiency processor that implements the Armv8-A architecture.
- ❑ The Cortex-A53 processor has one to four cores, each with an L1 memory system and a single shared L2 cache. It can be combined with other Cortex-A CPUs in a big. LITTLE configuration.

PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Processor



PROCESSOR SELECTION CRITERIA

Case study-1: Handheld Device

Processor

Architecture	Armv8-A
Multicore	1-4x Symmetrical Multiprocessing (SMP) within a single processor cluster, and multiple coherent SMP processor clusters through AMBA 4 technology
ISA Support	• AArch32 for full backward compatibility with Armv7 • AArch64 for 64-bit support and new architectural features • TrustZone security technology • Neon advanced SIMD • DSP & SIMD extensions • VFPv4 floating point • Hardware virtualization support

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

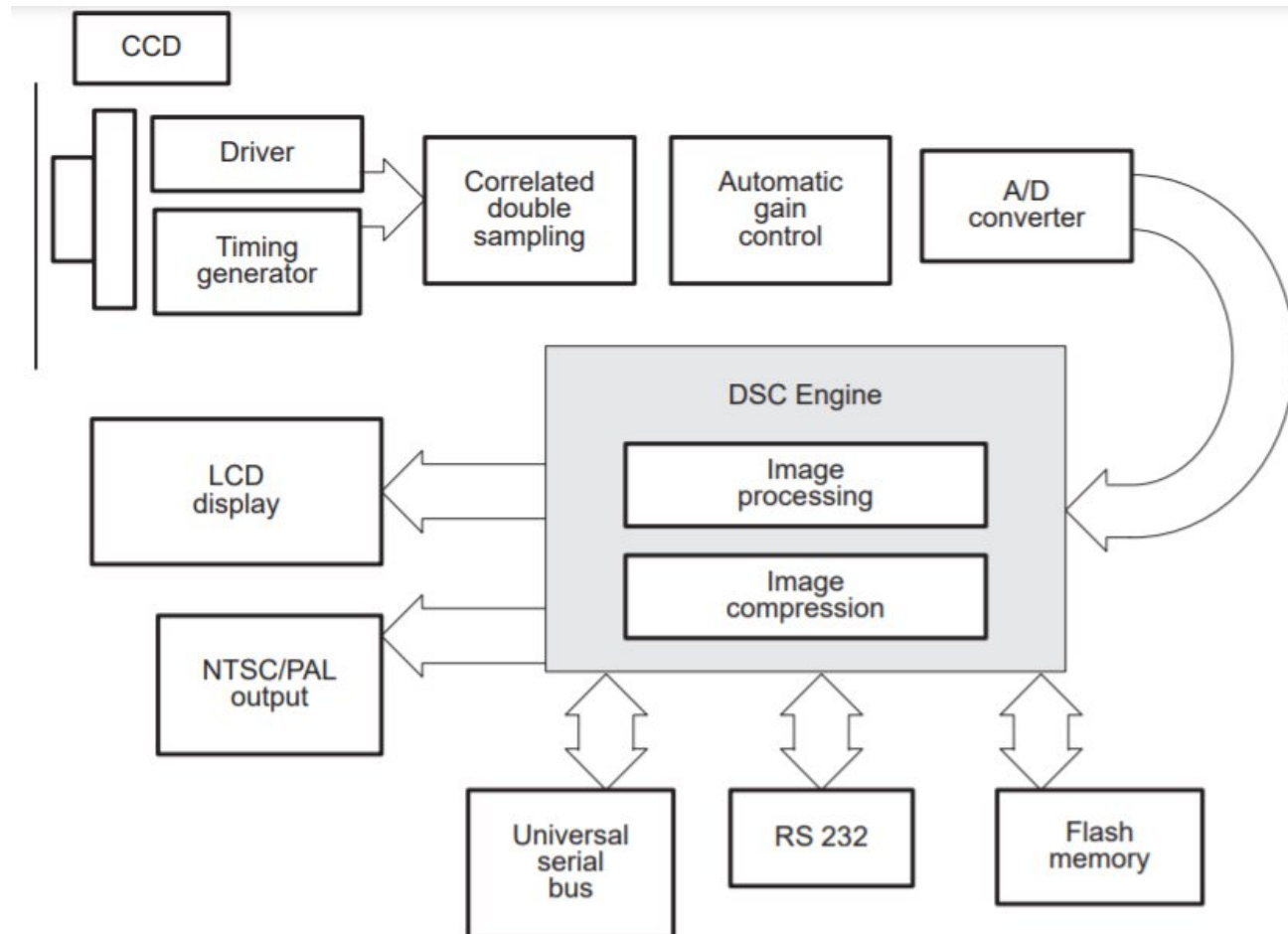
Digital Still Camera

- ❑ Digital still cameras require a significant amount of silicon contents, including the sensor (CCD or CMOS), the analog components (ADC, NTSC encoder, ...) and the engine (Digital Signal Processor).
- ❑ DSP - is the brain of the camera and is responsible for performing all the computations needed to process and compress the image.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Digital Still Camera



PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Digital Still Camera

- ❑ Most DSCs use a CCD imager to sense the images.
- ❑ The driver electronics and the Timing Generator circuitry generate the necessary signal to clock the CCD.
- ❑ Correlated Double Sampling and Automatic Gain Control electronics are used to get a good-quality image signal from the CCD sensor.
- ❑ This CCD data is then digitized and fed into the DSC engine.
- ❑ All the image-processing and image-compression operations are performed in the DSC engine.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

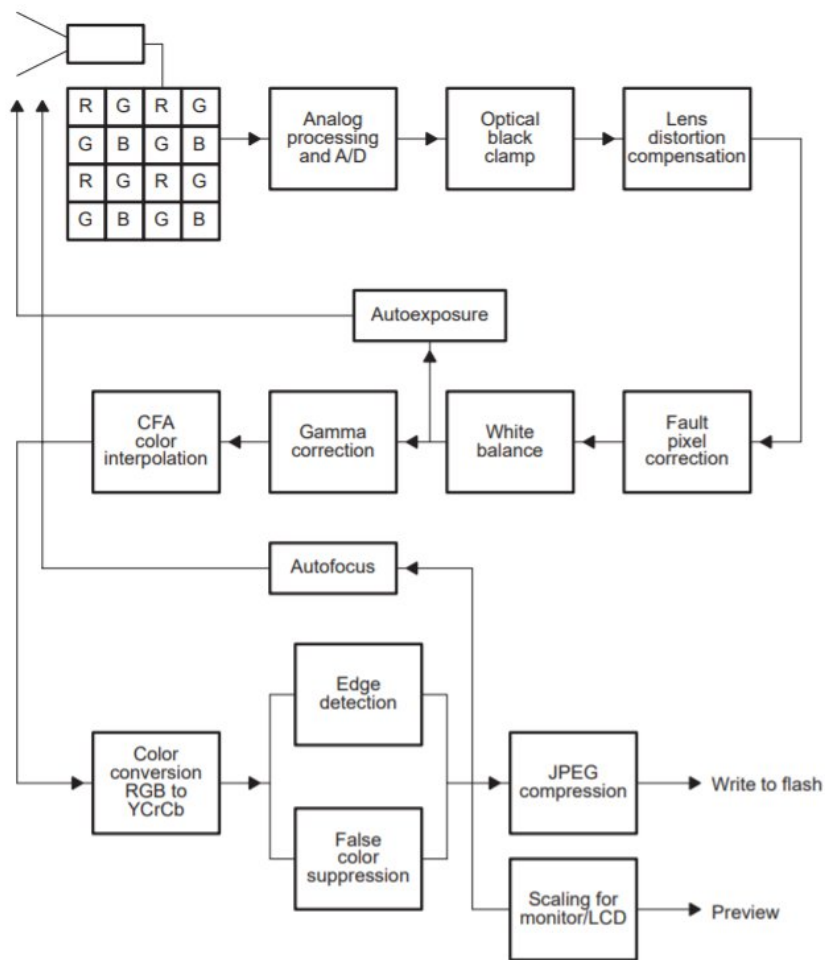
Digital Still Camera

- ☐ On most DSCs, the user has the ability to view the image to be captured on the LCD display.
- ☐ The compressed images are stored in Flash memory for later use.
- ☐ Most DSC systems also provide an NTSC/PAL video signal to view the captured images (also the preview images) on a TV monitor.
- ☐ The current DSCs also provide ways to connect to the external PC or printer through an RS-232 or a USB port.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Image Pipeline



PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Image Pipeline

- ❑ The Colour Filtered Array (CFA) data needs to undergo significant amount of image processing before the image can be finally presented in a usable format for compression.
- ❑ All these processing stages are collectively called the “image pipeline.”
- ❑ Most of these tasks are multiply-accumulate (MAC) intensive operations.
- ❑ The TMS320C54x DSP is well suited to perform these tasks efficiently and generate a high-quality image that is close to the image quality offered by traditional film from the raw CCD data.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Processor - TMS320C54x DSC

- ☐ This system interfaces to the CCD/CMOS module directly.
- ☐ The DSP reads the data from the sensor, processes the raw sensor data, and writes it to the SDRAM.
- ☐ The built-in NTSC/PAL encoder chip on the board allows direct display of the processed picture on the TV monitor.
- ☐ All image pipeline operations, including JPEG compression, can be performed on the TMS320C54x on this 16 x 16 block of pixels and then the compressed bit stream is written out to the SDRAM.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Processor - TMS320C54x DSC

- ❑ On this system, all image pipeline operations can be executed on chip since only a small 16 x 16 block of the image is used.
- ❑ The TMS320C549 is well suited due to its large on-chip memory (32K x 16-bit RAM and 16K x 16-bit ROM).
- ❑ In this way, the processing time is kept short, because there is no need for external high-speed memory.
- ❑ This device offers high performance (100 MIPS) at low power consumption (0.45mA/MIPS).

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Processor - TMS320C54x DSC

- ❑ Due to the efficiency of the TMS320C54x instruction set and architecture, the entire image pipeline, including JPEG, takes about 150 cycles/pixel.
- ❑ Hence, a 1-Mpixel CCD image can be processed in 1.5 seconds on a 100-MHz TMS320C54x.
- ❑ This offers about a 2-second shot-to-shot delay, including data movement from external memory to on-chip memory.

PROCESSOR SELECTION CRITERIA

Case study-2: Digital Camera

Processor - TMS320C54x DSC

Task	Cycles/Pixel
Preprocessing (Black Clamping, Gain, White Balance, Gamma Correction)	22
Color Space Conversion	10
Interpolation	41
Edge Enhancement & False Color Suppression	27
Total	90
4:1:1 Decimation & JPEG Encoding	62
Total	152

Memory	K Bytes
Program	1.7
Data	4.6

PROCESSOR SELECTION CRITERIA

Case study-3: ATM

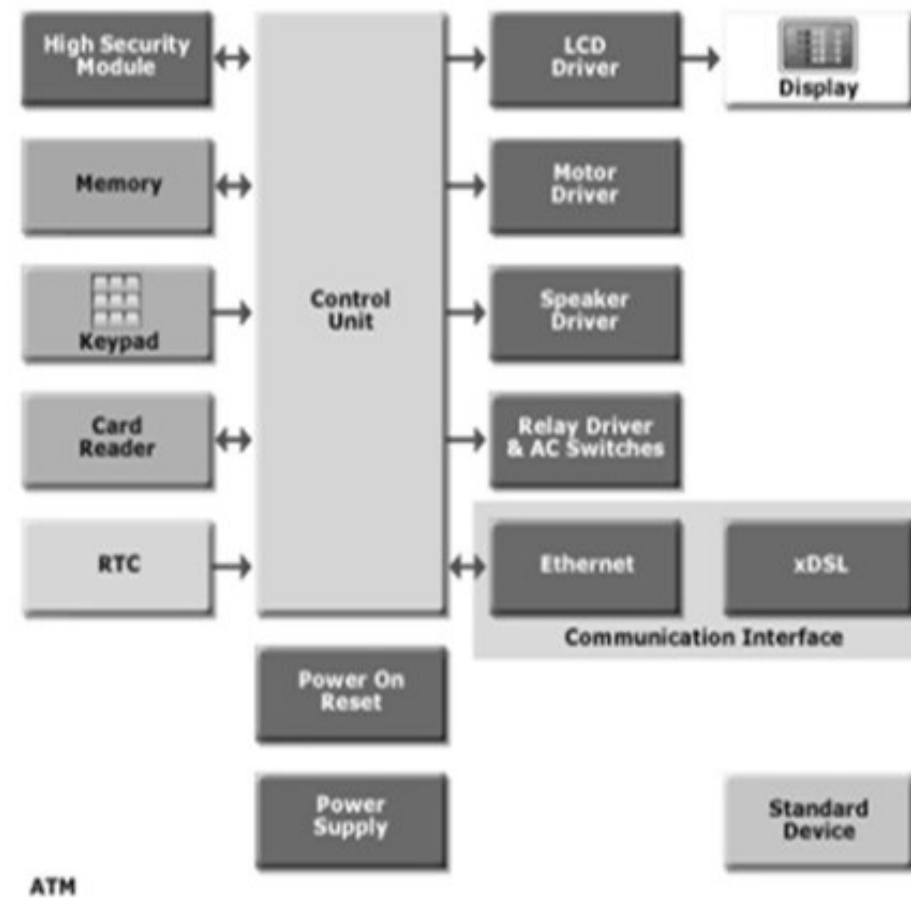
An automated teller machine or automatic teller machine (ATM) is a computerized telecommunications device that provides a financial institution's customers a secure method of performing financial transactions in a public space without the need for a human clerk or bank teller.



PROCESSOR SELECTION CRITERIA

Case study-3: ATM

ATM - Block Diagram



PROCESSOR SELECTION CRITERIA

Case study-3: ATM

ATM - Hardware

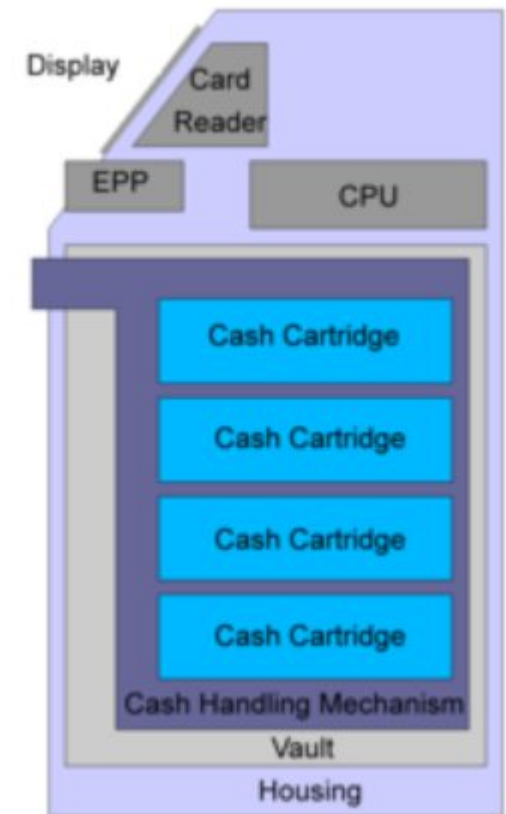
- ☐ CPU (to control the user interface and transaction devices)
- ☐ Magnetic and/or Chip card reader (to identify the customer)
- ☐ PIN Pad (similar in layout to a Touch tone or Calculator keypad), often manufactured as part of a secure enclosure.
- ☐ Secure cryptoprocessor, generally within a secure enclosure.
- ☐ Display (used by the customer for performing the transaction)
- ☐ Function key buttons (usually close to the display) or a Touchscreen (used to select the various aspects of the transaction)
- ☐ Record Printer (to provide the customer with a record of their transaction)
- ☐ Vault (to store the parts of the machinery requiring restricted access)
- ☐ Housing (for aesthetics and to attach signage to)

PROCESSOR SELECTION CRITERIA

Case study-3: ATM

ATM - Vaults

- ☐ Dispensing mechanism (to provide cash or other items of value)
- ☐ Deposit mechanism (to take items of value from the customer)
- ☐ Security sensors (Magnetic, Thermal, Seismic)
- ☐ Locks (to ensure controlled access to the contents of the vault)



PROCESSOR SELECTION CRITERIA

Case study-3: ATM

ATM - Networking

- ☐ The internet service provider (ISP) also plays an important role in the ATMs.
- ☐ This provides communication between ATM and host processors.
- ☐ When the transaction is made, the details are input by the cardholder.
- ☐ This information is passed on to the host processor by the ATM.
- ☐ The host processor checks these details with an authorized bank.
- ☐ If the details are matched, the host processor sends the approval code to the machine so that the cash can be transferred.)



PROCESSOR SELECTION CRITERIA

Special Purpose Processor

- ❑ Using a single-purpose processor in an embedded system results in several design metric benefits and drawbacks, which are essentially the inverse of those for general purpose processors.
- ❑ Adv: Performance may be fast, size and power may be small, and unit-cost may be low for large quantities
- ❑ Diadv: While design time and NRE costs may be high, flexibility is low, unit cost may be high for small quantities, and performance may not match general-purpose processors for some applications.